

Numerical Questions on Agile Metrics: Velocity, Throughput, and Cycle Time

1.

An Agile team completes the following story points in three consecutive sprints:

- Sprint 1: 35 story points
- Sprint 2: 42 story points
- Sprint 3: 38 story points

Calculate the **average velocity** of the team.

2.

A product backlog contains **200 story points**. The Agile team's average velocity is **40 story points per sprint**.

How many sprints are required to complete the backlog?

3. Throughput Calculation

A software maintenance team completes the following number of tickets in four weeks:

- Week 1: 18 tickets
- Week 2: 22 tickets
- Week 3: 25 tickets
- Week 4: 15 tickets

Calculate the **average throughput** of the team.

4.

A team has **150 pending support tickets**. The team's average throughput is **30 tickets per week**.

Estimate the time required to clear all pending tickets.

5.

A development team tracks the following tasks:

- Task T1 starts on Day 2 and is completed on Day 8.
- Task T2 starts on Day 4 and is completed on Day 10.
- Task T3 starts on Day 6 and is completed on Day 15.
- Task T4 starts on Day 9 and is completed on Day 12.

Calculate the **average cycle time** of the team.

6.

An Agile team working on an e-commerce application completes the following story points:

- Sprint 1: 45 story points
- Sprint 2: 50 story points
- Sprint 3: 55 story points

The product owner has a remaining backlog of **300 story points**.

During the same period, the team completes the following tickets:

- Week 1: 24 tickets
- Week 2: 30 tickets
- Week 3: 36 tickets

The average cycle time of completed tasks is **5 days**.

Calculate:

- a) Average velocity
- b) Number of sprints required to complete the backlog
- c) Average throughput
- d) Expected time required based on throughput

7.

A mobile application development team follows two-week sprints. The completed story points for the last four sprints are:

- Sprint 1: 32 story points
- Sprint 2: 40 story points
- Sprint 3: 36 story points
- Sprint 4: 44 story points

During the same period, the team completed:

- Sprint 1: 12 user stories
- Sprint 2: 15 user stories
- Sprint 3: 18 user stories
- Sprint 4: 20 user stories

The average cycle times for the completed stories were:

- Sprint 1: 4 days
- Sprint 2: 5 days
- Sprint 3: 6 days

- Sprint 4: 5 days

Calculate:

- a) Team velocity
- b) Throughput per sprint
- c) Average cycle time
- d) Estimate the number of sprints required if **250 story points** remain in the backlog.